

UYEN HUYNH

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OBJECTIVE

MLOps/AI Engineer with strong expertise in building end-to-end production pipelines, optimizing inference latency, and deploying scalable architectures using FastAPI and Docker. Proven track record of handling high-dimensional datasets and implementing Medallion Architecture for data-intensive applications.

TECHNICAL SKILLS

MLOps & Backend Engineering: FastAPI (Async/Await), Docker Containerization, Redis (Caching strategies), PostgreSQL, CI/CD (GitHub Actions), RESTful API Design, Microservices Architecture.

Machine Learning & Deep Learning: PyTorch, TensorFlow, Graph Neural Networks (GCN), LightGBM (Gradient Boosting), YOLOv8 (Object Detection), SHAP (Explainable AI), Feature Engineering (Imputation/Scaling), Scikit-learn, Medallion Architecture (Bronze/Silver/Gold Pipelines).

Generative AI & LLM Systems: LangChain (LCEL), RAG (Retrieval-Augmented Generation), Vector Databases (Qdrant/Chroma), Microsoft Presidio (PII Redaction), Prompt Engineering, Semantic Search.

Software Quality & Tools: Python 3.11+, Pytest (Unit/Integration Testing), Git/GitHub, SonarQube, Next.js (Basic integration).

KEY PROJECTS

- **MediAI - ICU Risk Prediction Platform V4** [\[GitHub\]](#) [\[Production\]](#)

Production-ready MLOps platform for ICU decision support — Personal Project

Tech Stack: Python, FastAPI, LightGBM V2, Redis, LangChain, Next.js 14, Docker, PostgreSQL

- Architected a microservices-based inference engine serving 70,000+ samples using **FastAPI** and **Redis**, reducing latency to **50ms** (10x improvement).
- Implemented a RAG pipeline with **LangChain** and Vector Search for real-time information retrieval, ensuring strictly enforced data privacy (RBAC/Encryption).
- Deployed fully containerized pipelines (Docker) with automated CI/CD workflows on Vercel/Render.

- **T-DDI: Robust Prediction of Drug Interactions using Chemical Descriptors** [\[Live Demo\]](#)

Ensemble ML and DL framework for Drug-Drug Interaction (DDI) using structured molecular descriptors — Team Research Project

Tech Stack: PyTorch, TabTransformer, QSAR Descriptors, SMILES, LIME, Focal Loss

- Developed a Transformer-based architecture to classify complex sequence interactions (800k+ pairs), achieving 93.7% accuracy.
- Designed a custom loss function (Focal Loss) to handle severe class imbalance in large-scale datasets.
- Engineered a feature extraction pipeline converting raw string sequences (SMILES) into high-dimensional vector representations.

- **DeepGPT-DILI - Multi-modal Graph & Sequence Analysis System** [\[Github\]](#) [\[Publication\]](#)

Ensemble Deep Learning framework for high-dimensional data classification — Team Research Project

Tech Stack: PyTorch, GCN, Attention-RNN, XGBoost, Docker, Sequence Processing

- Published at **MICCAI 2025** with State-of-the-Art performance (**0.921 AUROC**) by engineering a robust ensemble voting mechanism combining **Graph Neural Networks (GCN)**, **XGBoost**, and **Extra Trees**.
- Designed a multi-modal architecture leveraging orthogonal data representations: **Graph Structures** for topological analysis and **Attention-based RNNs** for sequential data processing.
- Implemented advanced feature engineering pipelines and dimensionality reduction via **PCA**, optimizing model performance for high-dimensional sparse datasets.

- **Additional Machine Learning Projects**

- Object Detection System: Built real-time helmet detection using YOLOv8 for safety monitoring application.

- Cardiovascular Prediction: Developed stacking ensemble model for risk assessment using multiple base learners.
- Traffic Sign Detection: Implemented computer vision pipeline for autonomous vehicle applications.
- Microbial Genome Analysis: Sequencing and analyzing microbial DNA data (team project).

WORKING EXPERIENCE

AI Researcher

5/2024 - Present

Taipei Medical University, Taipei City, Taiwan

- Developed **DeepGPT-DILI** (MICCAI 2025, 0.921 AUROC) and **T-DDI** (Nature Comms submitted), focusing on optimizing Graph Neural Networks (GCN) and Transformer architectures for complex sequence data.
- Engineered high-performance hybrid models combining Gradient Boosting and Deep Learning, significantly improving prediction accuracy over baseline systems.
- **Academic Service:** Verified Peer Reviewer for PLOS ONE (International Scientific Journal) - Evaluating novel research in applied AI systems.

AI Researcher

1/2023 - 5/2024

AI Vietnam, Ho Chi Minh City, Vietnam

- **VLA Team (LLM Research):** Co-authored the paper "*Curating Multi-Mode CoT for Efficient Math Reasoning*" (2025).
- Implemented **Distillation** and **Chain-of-Thought (CoT)** techniques to fine-tune Llama 3.2 3B for mathematical reasoning, boosting performance to **80.06%** on GSM8K.
- Engineered an automated **Data Curation Pipeline** to filter noise and standardize large-scale training datasets.
- **AIMA Team:** Built and deployed applied AI solutions, handling data preprocessing pipelines for large-scale diagnostic datasets.

Software Development Intern

7/2024 - 9/2024

Mat Bao Corporation, Tan Binh, Ho Chi Minh city, Vietnam

- Developed and maintained web applications and ensuring optimal performance.
- Worked with .NET framework and SQL databases for backend development tasks.

EDUCATION

Bachelor of Engineering in Biotechnology

expected 2026

University of Agriculture and Forestry, Ho Chi Minh City, Vietnam

- AIVIETNAM Member (2023 - Present): Active participation in AI/ML community.
- Relevant Coursework: Data Structures Algorithms, Database Systems, Machine Learning, Statistics.
- Microbial Genome Analysis: Team project on DNA sequencing and computational analysis.

LANGUAGE

- English: B2
- French: B2
- Chinese: HSK 4
- Vietnamese: Native